

Practical 1: SNP and STR Analysis

Bioinformatics and Statistical Genetics (BSG)

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```
install.packages("data.table")  
for mac do this first: install.packages("Rsolnp", type = "binary") ##  
install.packages("HardyWeinberg")
```

```
library(data.table)  
library(HardyWeinberg)
```

```
## Loading required package: mice
```

```
##
```

```
## Attaching package: 'mice'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      filter
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      cbind, rbind
```

```
## Loading required package: nnet
```

```
## Loading required package: Rsolnp
```

```
## Loading required package: shape
```

SNP dataset

Loading dataset

```
df <- read.table("TSICHR22RAW.raw", header = TRUE)  
genetic_info <- df[, 7:ncol(df)]  
geno_dt <- as.data.table(genetic_info)
```

1. How many variants are there in this database? What percentage of the data is missing?

```
n_variants <- ncol(geno_dt)

cat("Number of variants:", n_variants, "\n")

## Number of variants: 20649

pct_missing <- mean(is.na(geno_dt)) * 100
cat("Percentage missing:", round(pct_missing, 4), "%\n")
```

```
## Percentage missing: 0.1987 %
```

2. Calculate the percentage of monomorphic variants. Exclude all monomorphics from the database for all posterior computations of the practical. How many variants do remain in your database?

Monomorphic means no variation, so we look for columns with same value all across.

```
n_unique <- geno_dt[, sapply(.SD, function(x) uniqueN(na.omit(x)))]
monomorphic <- n_unique <= 1
cat("Monomorphic %:", mean(monomorphic) * 100, "\n")

## Monomorphic %: 11.45818

geno_poly <- geno_dt[, !monomorphic, with = FALSE]

n_poly <- ncol(geno_poly)
cat("Remaining variants:", n_poly, "\n", "Removed variants:", ncol(geno_dt)-n_poly, "\n", "These polymorphic variants were used for all subsequent analyses.")

## Remaining variants: 18283
## Removed variants: 2366
## These polymorphic variants were used for all subsequent analyses.
```

3. Report the genotype counts and the minor allele count of polymorphism rs8138488 C, and calculate the MAF of this variant.

```
geno_counts <- geno_poly[, .N, by = rs8138488_C]
print(geno_counts)
```

```
##      rs8138488_C      N
##      <int> <int>
## 1:         0    41
## 2:         1    47
## 3:         2    14
```

```

# genotype counts
g0 <- geno_poly[rs8138488_C == 0, .N]
g1 <- geno_poly[rs8138488_C == 1, .N]
g2 <- geno_poly[rs8138488_C == 2, .N]

# allele counts
n_ref <- 2 * g0 + g1
n_alt <- 2 * g2 + g1
n_total <- n_ref + n_alt

# allele frequencies
freq_ref <- n_ref / n_total
freq_alt <- n_alt / n_total

# MAF
MAF <- min(freq_ref, freq_alt)

cat(paste(
  "Genotype counts:",
  paste0(" 0 (AA): ", g0),
  paste0(" 1 (AB): ", g1),
  paste0(" 2 (BB): ", g2),
  "Allele counts:",
  paste0(" Reference: ", n_ref),
  paste0(" Alternate: ", n_alt),
  "Allele frequencies:",
  paste0(" Reference: ", round(freq_ref, 4)),
  paste0(" Alternate: ", round(freq_alt, 4)),
  paste0("MAF: ", round(MAF, 4)),
  sep = "\n"
))

```

```

## Genotype counts:
##   0 (AA): 41
##   1 (AB): 47
##   2 (BB): 14
## Allele counts:
## Reference: 129
## Alternate: 75
## Allele frequencies:
## Reference: 0.6324
## Alternate: 0.3676
## MAF: 0.3676

```

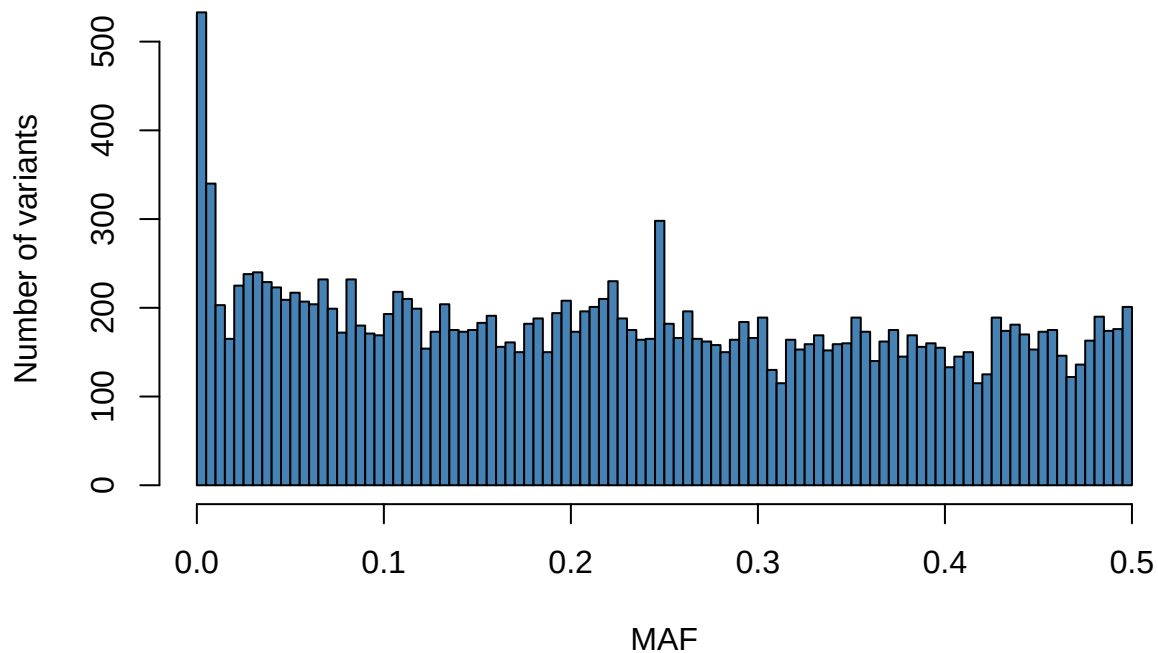
4. Compute the minor allele frequencies (MAF) for all markers, and make a histogram of it. Does the MAF follow a uniform distribution? What percentage of the markers have a MAF below 0.05? And below 0.01? Can you explain the observed pattern? Uniform distribution at $[0, 0.5]$. The range from 0 to 0.5 can be easily explained due to the fact that a frequency higher than 0.5 would indicate this is not the minor allele.

```
# Function to compute MAF for one column
compute_maf <- function(x) {
  g0 <- sum(x == 0, na.rm = TRUE)
  g1 <- sum(x == 1, na.rm = TRUE)
  g2 <- sum(x == 2, na.rm = TRUE)
  n_ref <- 2 * g0 + g1
  n_alt <- 2 * g2 + g1
  n_total <- n_ref + n_alt
  if (n_total == 0) return(NA)
  min(n_ref, n_alt) / n_total
}

# Apply to all columns
maf <- sapply(geno_poly, compute_maf)

hist(maf,
      breaks = 100,
      col = "steelblue",
      main = "Distribution of Minor Allele Frequencies",
      xlab = "MAF",
      ylab = "Number of variants")
```

Distribution of Minor Allele Frequencies



```
# We can visually assess, but also Kolmogorov-Smirnov test against uniform too
ks.test(maf, "punif", min = 0, max = 0.5)
```

```
## Warning in ks.test(maf, "punif", min = 0, max = 0.5): ties should not be
## present for the Kolmogorov-Smirnov test
```

```
##
## One-sample Kolmogorov-Smirnov test
##
## data: maf
## D = 0.06309, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

```
# Below 0.05
pct_below_005 <- mean(maf < 0.05, na.rm = TRUE) * 100
cat("Percentage with MAF < 0.05:", round(pct_below_005, 2), "%\n")
```

```
## Percentage with MAF < 0.05: 14.23 %
```

```
# Below 0.01
pct_below_001 <- mean(maf < 0.01, na.rm = TRUE) * 100
cat("Percentage with MAF < 0.01:", round(pct_below_001, 2), "%\n")
```

```
## Percentage with MAF < 0.01: 4.73 %
```

Answers: A Kolmogorov-Smirnov test against a uniform distribution rejects uniformity ($p < 0.001$), consistent with the visual L-shape. It is true that this test should be interpreted with caution because MAF values are discrete and the sample size is large, so even small deviations from uniformity could be statistically significant.

14.23 % of the variants have MAF lower than 0.05 and 4.73 % MAF lower than 0.01. What this tells us is that in this population rare variants are over-represented, and that the very rare end is actually more densely populated. Nevertheless, most markers (about 86%) are common, with $MAF \geq 0.05$.

Upon reflection and consulting possible explanations, we learn that this pattern is expected from population genetics. Every variant arises as a single mutation and so starts at the lowest possible frequency. Then, most are lost through genetic drift before becoming common, and human population growth adds a further excess of young, rare variants.

5. Calculate the observed heterozygosity H_0 , and make a histogram of it. What is, theoretically, the range of variation of this statistic?

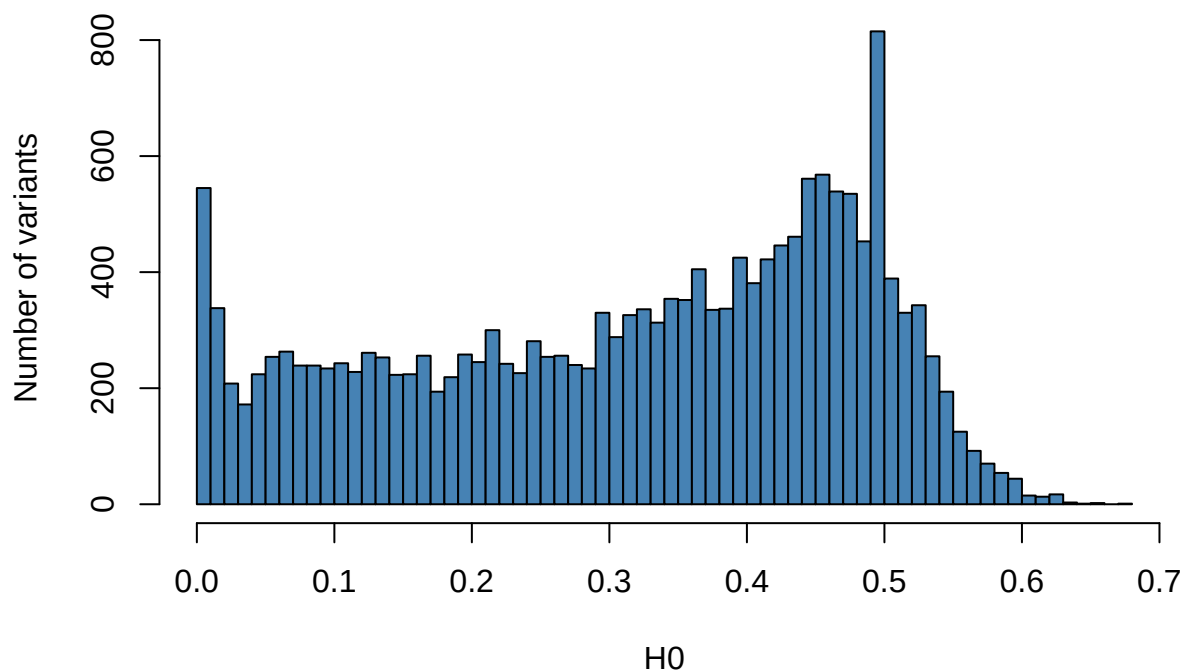
At a given SNP, we have two chromosome copies of each parent. If those two copies have different alleles, it is heterozygous. So code 1 = heterozygous.

```
# Function to compute H0 for one column
compute_h0 <- function(x) {
  x <- x[!is.na(x)]
  if (length(x) == 0) return(NA)
  mean(x == 1)
}

h0 <- geno_poly[, lapply(.SD, function(x) mean(x == 1, na.rm = TRUE))]
h0 <- unlist(h0)

hist(h0,
     breaks = 50,
     col = "steelblue",
     main = "Distribution of Observed Heterozygosity (H0)",
     xlab = "H0",
     ylab = "Number of variants")
```

Distribution of Observed Heterozygosity (H_O)



Answer:

The observed heterozygosity (H_O) is the proportion of individuals with a heterozygous genotype. Since it is a proportion, its theoretical range is **0 to 1**, where 0 means that no individuals are heterozygous and 1 means that all individuals are heterozygous.

So:

$$0 \leq H_O \leq 1$$

In practice, we are seeing values much below 1 because it is rare for every individual in a sample to be heterozygous.

Compute for each marker its expected heterozygosity (H_e). Make a histogram of the expected heterozygosity. What is, theoretically, the range of variation of this statistic? What is the average of H_e for this database?

```
# Function to compute He for one variant
compute_he <- function(x) {
  x <- x[!is.na(x)]
  g0 <- sum(x == 0)
  g1 <- sum(x == 1)
  g2 <- sum(x == 2)

  n_ref <- 2 * g0 + g1
```

```

n_alt <- 2 * g2 + g1
n_total <- n_ref + n_alt

if (n_total == 0) return(NA)

p_ref <- n_ref / n_total
p_alt <- n_alt / n_total

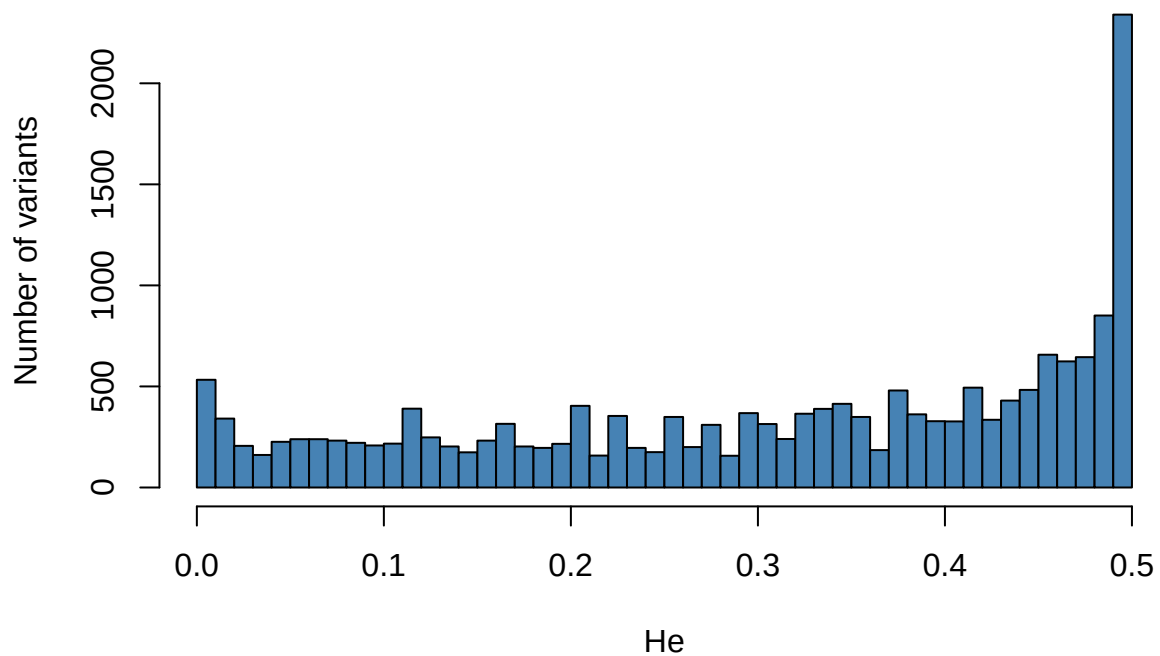
1 - (p_ref^2 + p_alt^2)
}

he <- geno_poly[, lapply(.SD, compute_he)]
he <- unlist(he)

hist(he,
     breaks = 50,
     col = "steelblue",
     main = "Expected Heterozygosity (He)",
     xlab = "He",
     ylab = "Number of variants")

```

Expected Heterozygosity (He)



```

cat(paste(
  paste0("Min He: ", min(he, na.rm = TRUE)),
  paste0("Max He: ", max(he, na.rm = TRUE)),
  paste0("Mean He: ", mean(he, na.rm = TRUE)),

```

```
sep = "\n"
))
```

```
## Min He: 0.00975586312956556
## Max He: 0.5
## Mean He: 0.311584071584143
```

Answers: For a bi-allelic marker, the maximum occurs when both alleles have frequency 0.5:

$$H_e = 1 - (0.5^2 + 0.5^2) = 0.5$$

The minimum occurs when one allele has frequency 1 and the other has frequency 0:

$$H_e = 1 - (1^2 + 0^2) = 0$$

Therefore:

$$0 \leq H_e \leq 0.5$$

The maximum occurs when both alleles are equally frequent. In this database, the average H_e is 0.3116.

STR dataset

```
data(NistSTRs)

# Save to a data frame
str_df <- as.data.frame(NistSTRs)
```

1. How many individuals and how many STRs contains the database?

```
cat("Individuals:", nrow(str_df), "\n")
```

```
## Individuals: 361
```

```
cat("STRs:", ncol(str_df) / 2, "\n")
```

```
## STRs: 29
```

2. Write a function that determines the number of alleles for a STR. Determine the number of alleles for each STR in the database. Compute basic descriptive statistics of the number of alleles (mean, standard deviation, median, minimum, maximum).

```

count_alleles <- function(allele1, allele2) {
  all_alleles <- c(allele1, allele2)
  all_alleles <- all_alleles[!is.na(all_alleles)]
  length(unique(all_alleles))
}

str_names <- unique(sub("-1$|-2$", "", colnames(str_df)))

n_alleles <- sapply(str_names, function(s) {
  count_alleles(str_df[[paste0(s, "-1")]], str_df[[paste0(s, "-2")]])
})

cat(paste(
  "Number of alleles per STR:",
  paste0(names(n_alleles), ": ", n_alleles, collapse = "\n"),
  "",
  paste0("Mean: ", round(mean(n_alleles), 2)),
  paste0("SD: ", round(sd(n_alleles), 2)),
  paste0("Median: ", median(n_alleles)),
  paste0("Min: ", min(n_alleles)),
  paste0("Max: ", max(n_alleles)),
  sep = "\n"
))

```

```

## Number of alleles per STR:
## CSF1P0: 7
## D10S1248: 9
## D12S391: 16
## D13S317: 8
## D16S539: 7
## D18S51: 15
## D19S433: 15
## D1S1656: 15
## D21S11: 16
## D22S1045: 8
## D2S1338: 12
## D2S441: 11
## D3S1358: 9
## D5S818: 9
## D6S1043: 14
## D7S820: 9
## D8S1179: 10
## F13A01: 12
## F13B: 6
## FESFPS: 7
## FGA: 14
## LPL: 8
## Penta_C: 10
## Penta_D: 13
## Penta_E: 19
## SE33: 39
## TH01: 8

```

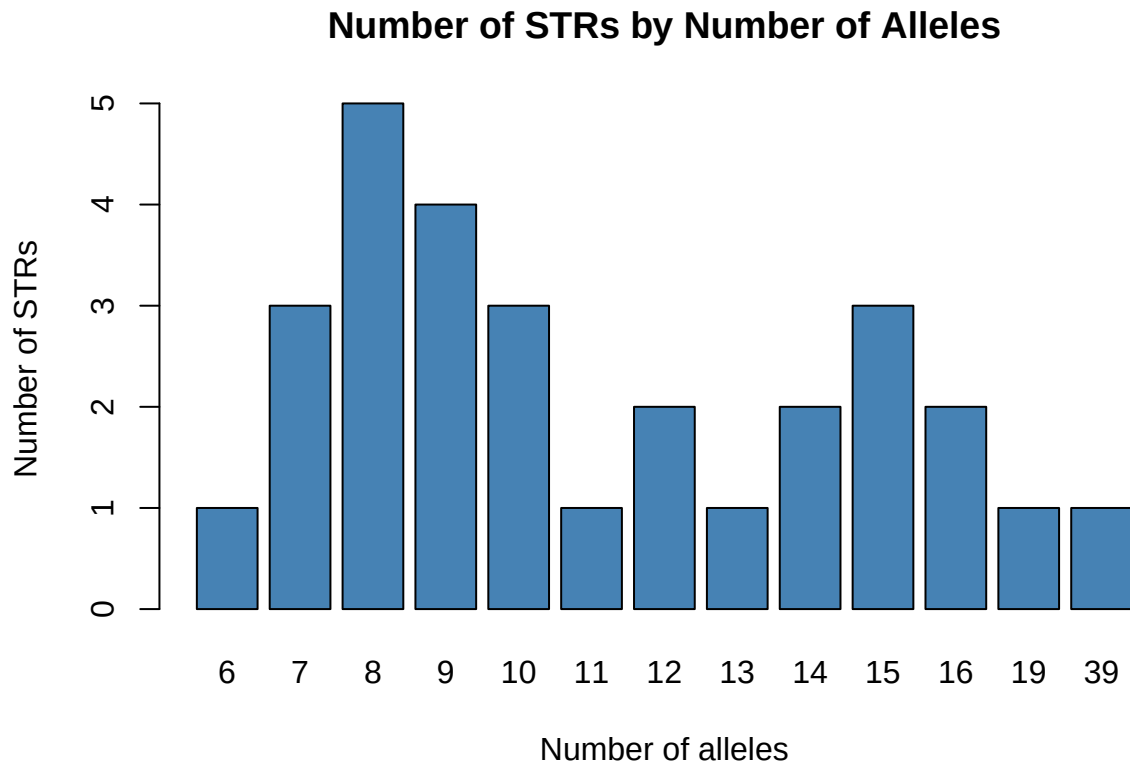
```
## TPOX: 8
## vWA: 10
##
## Mean: 11.86
## SD: 6.23
## Median: 10
## Min: 6
## Max: 39
```

3. Make a table with the number of STRs for a given number of alleles and present a barplot of the number STRs in each category. What is the most common number of alleles for an STR?

```
allele_table <- table(n_alleles)
print(allele_table)
```

```
## n_alleles
##  6  7  8  9 10 11 12 13 14 15 16 19 39
##  1  3  5  4  3  1  2  1  2  3  2  1  1
```

```
barplot(allele_table,
        main = "Number of STRs by Number of Alleles",
        xlab = "Number of alleles",
        ylab = "Number of STRs",
        col = "steelblue")
```



```
most_common <- as.numeric(names(allele_table)[which.max(allele_table)])
cat("Most common number of alleles:", most_common, "\n")
```

```
## Most common number of alleles: 8
```

4. Compute the expected heterozygosity for each STR. Make a histogram of the expected heterozygosity over all STRs. Compute the average expected heterozygosity over all STRs.

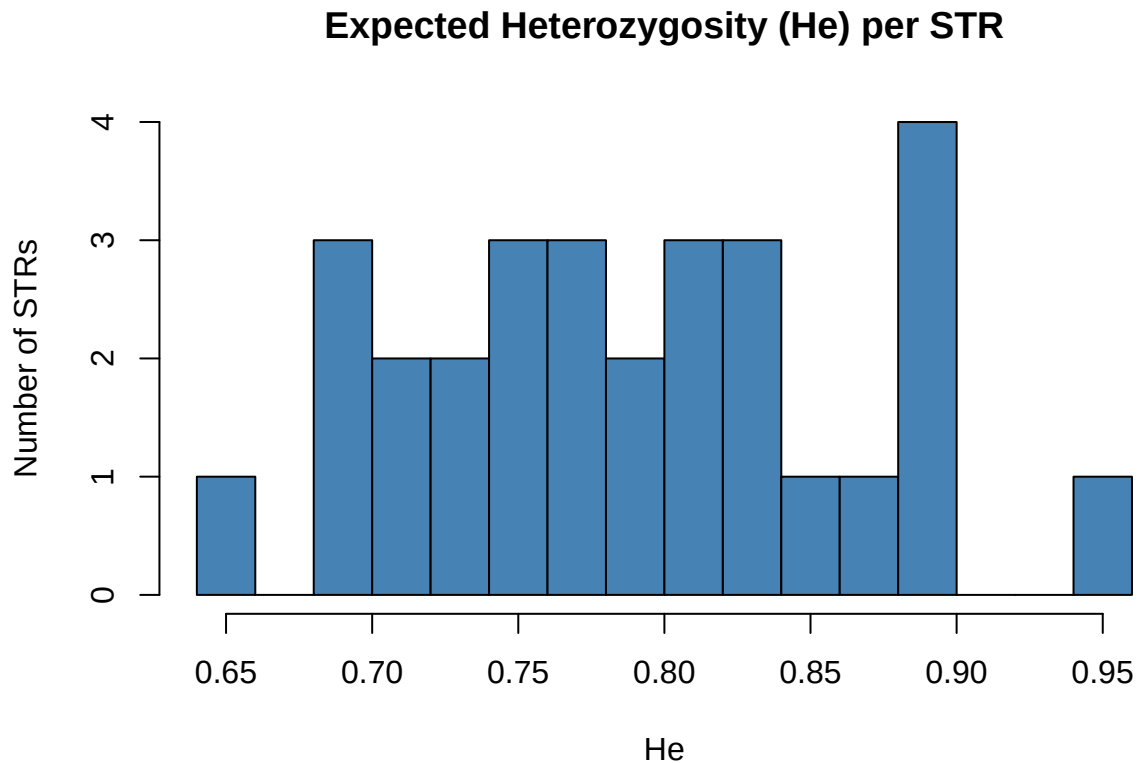
```
compute_he_str <- function(allele1, allele2) {
  all_alleles <- c(allele1, allele2)
  all_alleles <- all_alleles[!is.na(all_alleles)]
  freqs <- table(all_alleles) / length(all_alleles)
  1 - sum(freqs^2)
}

str_names <- unique(sub("-1$|-2$", "", colnames(str_df)))

he_str <- sapply(str_names, function(s) {
  compute_he_str(str_df[[paste0(s, "-1")]], str_df[[paste0(s, "-2")]])
})

hist(he_str, breaks = 15, col = "steelblue",
```

```
main = "Expected Heterozygosity (He) per STR",
xlab = "He", ylab = "Number of STRs")
```



```
cat("Average He:", round(mean(he_str), 4), "\n")
```

```
## Average He: 0.7904
```

5. Calculate also the observed heterozygosity for each STR. Plot observed against expected heterozygosity, using all STRs. What do you observe?

```
compute_h0_str <- function(allele1, allele2) {
  valid <- !is.na(allele1) & !is.na(allele2)
  mean(allele1[valid] != allele2[valid])
}

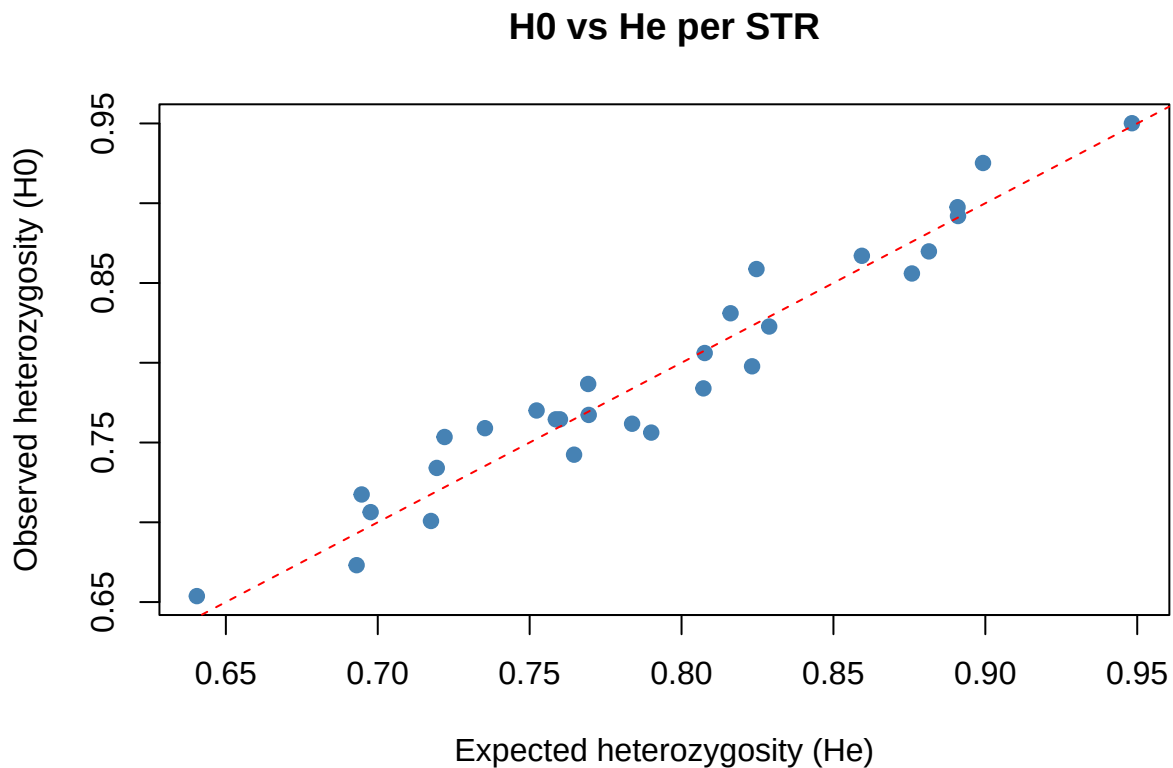
h0_str <- sapply(str_names, function(s) {
  compute_h0_str(str_df[[paste0(s, "-1")]], str_df[[paste0(s, "-2")]])
})

plot(he_str, h0_str,
```

```

xlab = "Expected heterozygosity (He)",
ylab = "Observed heterozygosity (H0)",
main = "H0 vs He per STR",
pch = 19, col = "steelblue")
abline(0, 1, col = "red", lty = 2)

```



```

fis <- (he_str - h0_str) / he_str
cat("Mean Fis:", round(mean(fis, na.rm = TRUE), 4), "\n")

```

```
## Mean Fis: -0.0024
```

Answer: Most points fall near the diagonal.

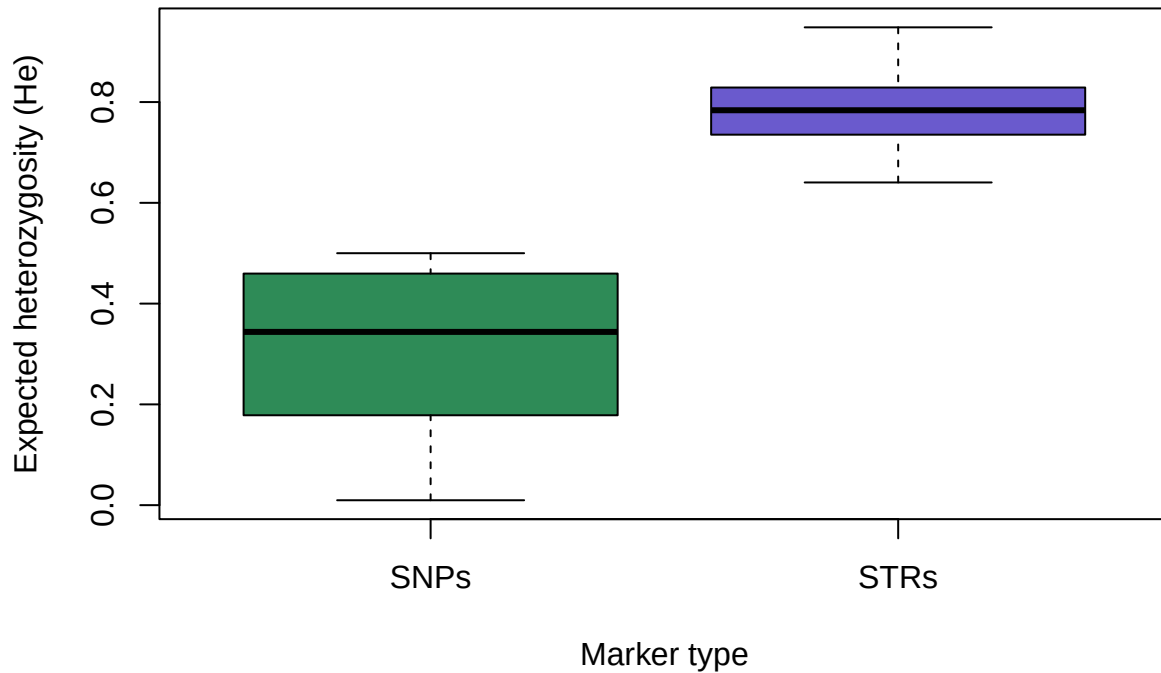
6. Compare, overall, the results you obtained for the SNP database with those you obtained for the STR database. What differences do you observe between these two types of genetic markers?

```

boxplot(list(SNPs = he, STRs = he_str),
  col = c("seagreen", "slateblue"),
  main = "Expected heterozygosity: SNPs vs STRs",
  xlab = "Marker type",
  ylab = "Expected heterozygosity (He)")

```

Expected heterozygosity: SNPs vs STRs



Answer: The SNPs has more than 18000 markers, and show a mean expected heterozygosity of only 0.31 and a mix of rare and common variants. The STR database, by contrast, contains just 29 markers but each one is highly multi-allelic. This allows STRs to reach much higher heterozygosity levels, with a mean expected heterozygosity of 0.79.

Both databases show observed heterozygosity very close to expected, with mean values near zero, indicating that both are approximately in equilibrium. However, the variability of heterozygosity differs in each: SNPs show a std of ~ 0.16 and STRs ~ 0.07 . This indicates that SNPs mutate slowly and accumulate rare variants, while STRs mutate rapidly and maintain many alleles at similar frequencies.